

EE400 [DIS] Answers

Problem 1, Part e:

The answer is yes. I was unsure how to do this part, so I had ChatGPT write me a script to further optimize the models beyond the methods that we have learned in class. My findings are as follows.

The INT8 TFLite Model reported to have a size of 5.75 KB, which was the smallest model that we produced. This could further be reduced by compressing the model. The code that was provided to me did this by combining the techniques of knowledge distillation and post-training quantization. Firstly, a smaller student model was trained to mimic the predictions of the larger teacher model which helped to retain high accuracy with fewer parameters. Then, by applying INT8 quantization, we reduced the precision of the model weights and activations from 32-bit floats to 8-bit integers, the process of quantization. This drastically shrunk the file with minimal loss of performance, as we can see from the results below:

INT8 Quant Model: 0.98 accuracy, 54 support at 5.75 KB

New Model: 0.96 accuracy, 54 support at 3.66 KB

As we can see, sacrificing 0.02 accuracy for 2.09 KB of size, which proves that the model can be shrunk further without sacrificing major performance.

Problem 2:

1. Types of learning blocks with examples:

- a. Supervised Learning Classification: Keras classification block for sensor data using a Keras neural network.
- b. Supervised Learning Regression: Keras regression block for continuous target data prediction like temperature forecasting or stock market data.
- c. Unsupervised learning anomaly detection: Visual anomaly detection (FOMO-AD) for detecting unusual patterns in image data.
- d. Supervised Learning Object Detection and Keyword Spotting: As the name suggests, wake word detection or FOV object detection given a visual input.
- e. Classical ML: I don't understand this one, but apparently its algorithms that use blocks like random forests, SVMs or K-NN.

2. Layers Under Classifier

- a. Input layer: defines the shape of incoming data
- b. Dense: Core computation layer for feature extraction and classification
- c. Reshape: Alters the tensor dimensions, for example flattening sequences
- d. Flatten: This converts multidimensional data into 1D before Dense Layers
- e. Dropout: Randomly disables neurons during training to reduce overfitting and over reliability
- f. 1D Convolution and Pooling: For temporal or sensor signals
- g. 2D convolution and Pooling: For image data such as image classification and object detection
- h. Output Layer: Provides model predictions

3. Model Compression in Deployment

- a. Quantization
- b. Unoptimized
- c. EON Compiler
- d. TFLite vs EON Compiler option

4. Synthetic Data Input Modalities

- a. Images
- b. Speech or audio
- c. Sound effects
- d. Synthetic data is important because it fills the gaps and simulates rare events that the test data may miss by providing cost effective yet diverse training examples. It can also overcome privacy or collection limitations on data by artificially filling in the gaps. Overall, it makes models more suited for the tasks they are trained for and improves its performance.